

10.7.90

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Question

A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$, where O is the origin. The circle contains the point $(-10, 2)$ in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$.

Determine the equation of the circle.

Theoretical solution

The tangent line is $y = x$, or $x - y = 0$. A point $\mathbf{P}$ on this line is represented parametrically.

$$\mathbf{P} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.1)$$

The distance from the origin is $\|\mathbf{P}\| = 4\sqrt{2}$.

$$\begin{aligned} |k| \left\| \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\| &= |k| \sqrt{2} = 4\sqrt{2} \\ \implies |k| &= 4 \end{aligned} \quad (0.2)$$

The two possible points are $\mathbf{P}_1 = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{P}_2 = -4 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Theoretical solution

Let the center be $\mathbf{C} = \begin{pmatrix} h \\ k \end{pmatrix}$. $\mathbf{P}$ is the foot of the perpendicular from $\mathbf{C}$ to the line $\mathbf{n}^\top \mathbf{x} = 0$ where $\mathbf{n} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$\mathbf{P} = \mathbf{C} - \frac{\mathbf{n}^\top \mathbf{C}}{\|\mathbf{n}\|^2} \mathbf{n} = \begin{pmatrix} h \\ k \end{pmatrix} - \frac{h - k}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{h + k}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.3)$$

Comparing this with the two possibilities for $\mathbf{P}$:

$$\text{Case 1: } \frac{h + k}{2} = 4 \implies h + k = 8 \quad (0.4)$$

$$\text{Case 2: } \frac{h + k}{2} = -4 \implies h + k = -8 \quad (0.5)$$

Theoretical solution

The chord lies on $x + y = 0$. The distance from $\mathbf{C}$ to this line is d_{chord} .

$$d_{chord} = \frac{\|h + k\|}{\sqrt{1^2 + 1^2}} = \frac{\|h + k\|}{\sqrt{2}} \quad (0.6)$$

With half-chord length $L = 3\sqrt{2}$, the radius r is given by:

$$r^2 = d_{chord}^2 + L^2 = \frac{(h + k)^2}{2} + 18 \quad (0.7)$$

The radius is also $r = \|\mathbf{C} - \mathbf{P}\|$.

$$r^2 = \left\| \frac{1}{2} \begin{pmatrix} h - k \\ k - h \end{pmatrix} \right\|^2 = \frac{(h - k)^2}{2} \quad (0.8)$$

Theoretical solution

Equating the two expressions for r^2 :

$$\begin{aligned}\frac{(h-k)^2}{2} &= \frac{(h+k)^2}{2} + 18 \\ (h-k)^2 &= (h+k)^2 + 36 \implies -4hk = 36 \implies hk = -9\end{aligned}\quad (0.9)$$

Now we solve for the center in each case.

For Case 1, $h+k=8$ and $hk=-9$.

$$h(8-h) = -9 \implies h^2 - 8h - 9 = 0 \implies (h-9)(h+1) = 0 \quad (0.10)$$

This gives centers $\mathbf{C}_A = \begin{pmatrix} 9 \\ -1 \end{pmatrix}$ and $\mathbf{C}_B = \begin{pmatrix} -1 \\ 9 \end{pmatrix}$.

Theoretical solution

For Case 2, $h + k = -8$ and $hk = -9$.

$$h(-8 - h) = -9 \implies h^2 + 8h - 9 = 0 \implies (h + 9)(h - 1) = 0 \quad (0.11)$$

This gives centers $\mathbf{C}_C = \begin{pmatrix} 1 \\ -9 \end{pmatrix}$ and $\mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

For any of these centers, the radius squared is $r^2 = \frac{(h-k)^2}{2} = \frac{(\pm 10)^2}{2} = 50$.

Theoretical solution

The point $\mathbf{Q} = \begin{pmatrix} -10 \\ 2 \end{pmatrix}$ is in the circle's interior, so we check $\|\mathbf{Q} - \mathbf{C}\|^2 < 50$.

$$\|\mathbf{Q} - \mathbf{C}_A\|^2 = \left\| \begin{pmatrix} -19 \\ 3 \end{pmatrix} \right\|^2 = 370 > 50 \quad (0.12)$$

$$\|\mathbf{Q} - \mathbf{C}_B\|^2 = \left\| \begin{pmatrix} -9 \\ -7 \end{pmatrix} \right\|^2 = 130 > 50 \quad (0.13)$$

$$\|\mathbf{Q} - \mathbf{C}_C\|^2 = \left\| \begin{pmatrix} -11 \\ 11 \end{pmatrix} \right\|^2 = 242 > 50 \quad (0.14)$$

$$\|\mathbf{Q} - \mathbf{C}_D\|^2 = \left\| \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\|^2 = 2 < 50 \quad (0.15)$$

The correct center is $\mathbf{C}_D = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$.

Theoretical solution

The equation of the circle is:

$$(x + 9)^2 + (y - 1)^2 = 50 \quad (0.16)$$

Expanding this gives:

$$x^2 + y^2 + 18x - 2y + 32 = 0 \quad (0.17)$$

In the matrix form $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$:

$$(x \ y) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} 9 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 32 = 0 \quad (0.18)$$

Theoretical solution

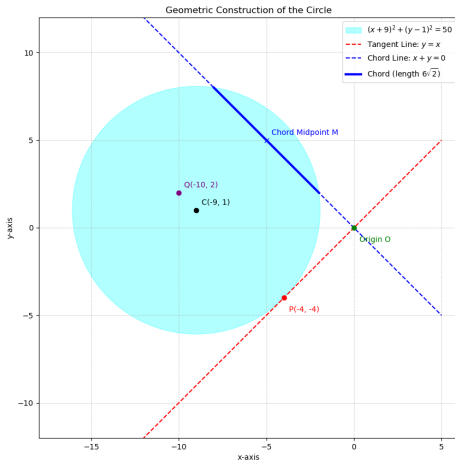


Fig :